

Amrit Gopinath

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RESEARCH INTERESTS

Natural Language Processing, Multilingual Representation Learning, Mechanistic Interpretability, Mixture-of-Experts Language Models, Classical Tamil NLP, Synthetic Media Detection

EDUCATION

Sri Sivasubramaniya Nadar College of Engineering, Chennai

2024 – Present

B.E. Computer Science and Engineering | CGPA: **9.383/10.0** up to 4th semester | Ranked **1st** in Department

RESEARCH OUTPUT

A Declarative–Procedural Perspective on Expert Routing in Bilingual Mixture-of-Experts Language Models 2026

Under anonymous review (ACL Rolling Review, May 2026 cycle); intended for AACL 2026 | First author

- ▶ Studied linguistic specialization and expert routing behaviour in bilingual English–German Mixture-of-Experts Transformers.
- ▶ Used probing methodologies and information-theoretic analyses to examine how expert routing relates to linguistic structure.
- ▶ Led the research design, experimentation, writing, and submission process.

MediaEval 2025 Working Notes: Synthetic Image Detection and Segmentation

2025

Accepted working-notes paper | First author

- ▶ Developed systems for distinguishing AI-generated and authentic imagery in a shared-task evaluation setting.
- ▶ Contributed to model design, experimentation, evaluation, and paper writing.

Exploring Swin Transformers, Luminance Input, and Hybrid Architectures for Synthetic Image Detection 2026

MediaEval 2026 Synthetic Image Detection submission | First author

- ▶ Compared pretrained Swin Transformer systems, luminance-only variants, MIL-style aggregation, and compact CNN/ResNet–GRU–ViT hybrid architectures.
- ▶ Evaluated constrained and open training settings using official metrics including accuracy, precision, recall, F1, AUC, AP, and confusion matrices.
- ▶ Best open-run system achieved an F1 score of **0.6997**.

FinMemeval 2026 Shared Task Submissions

2026

CLEF 2026 Working Notes — Accepted with Revisions | Co-author

- ▶ Contributed to three Financial NLP and multimodal benchmark submissions.
- ▶ Achieved Task 1 leaderboard results of **6th** place in English, **6th** place in Chinese, and **7th** place in Arabic.
- ▶ Worked on system development, experimentation, evaluation, and technical reporting.

RESEARCH EXPERIENCE

NLP Research Intern

2026 – Present

National Institute of Technology Tiruchirappalli

- ▶ Working on NLP research involving representation learning, linguistic structure, and evaluation of neural language models.
- ▶ Exploring sentence representations, probing methods, and multilingual/classical-language analysis.
- ▶ Studying computational approaches for classical Tamil linguistic resources, including verse–commentary relations, clause structure, finite and non-finite predicates, and syntax-aware retrieval.

TECHNICAL PROJECTS

CASSIAN – Graph-Augmented Code Understanding Platform

- ▶ Built a graph-augmented code retrieval system for structurally grounded code understanding.
- ▶ Generated function-level dependency graphs to support architectural reasoning over codebases.
- ▶ Developed an interactive D3-based blast-radius visualization and relevance-ranking pipeline under LLM context-window constraints.

A-Eye – Assistive Vision System

- ▶ Built a prototype combining object detection and contextual audio feedback for spatial-awareness assistance.
- ▶ Proposed a dual-camera vision-language system for real-time environmental understanding.
- ▶ Received institutional student research funding approval.

ACADEMIC SERVICE

Reviewer / Sub-reviewer, CLEF 2026 Working Notes 2026

- ▶ Reviewed assigned working-notes papers and provided technical feedback on shared-task submissions.

TECHNICAL SKILLS

Programming	Python, C, TypeScript, Java, LaTeX, Git
Machine Learning	PyTorch, Transformers, CNNs, Vision Transformers, Transfer Learning, Model Evaluation
NLP	Representation Learning, Probing, Multilingual Modeling, Mixture-of-Experts, Tokenization
Research Practice	Shared Tasks, Ablation Studies, Error Analysis, Technical Writing, Reproducible Experiments
CS Fundamentals	Data Structures & Algorithms, Object-Oriented Programming, DBMS

SELECTED ACHIEVEMENTS

- ▶ **FinMemeval 2026** – Top-10 multilingual leaderboard results in English, Chinese, and Arabic tracks.
- ▶ **Merit Scholarship** – 3rd Position, First Year, SSN College of Engineering.
- ▶ **Top 10** – GDG Hackathon.